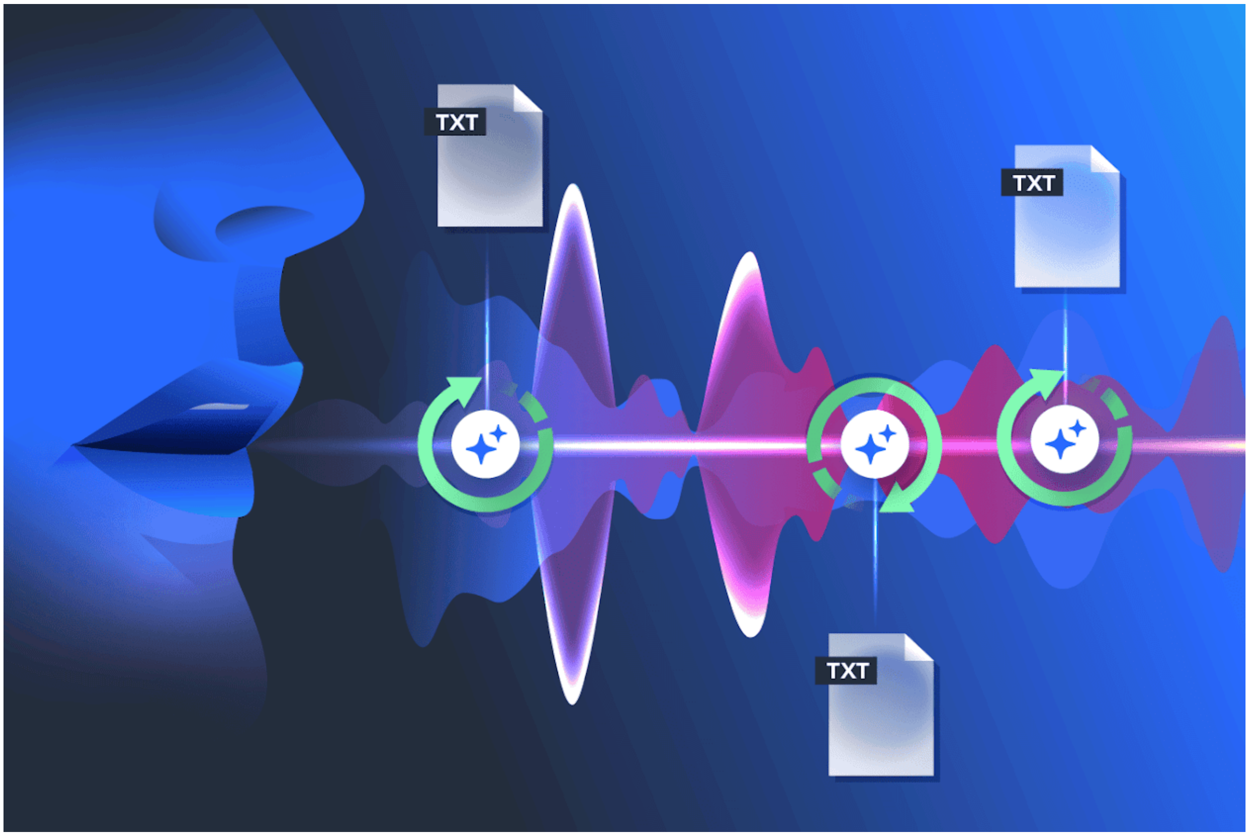

Burmese ASR using Kaldi & DNNs

12.09.2026

Assignment-4: ASR-Mini-Project

Group-4

AIE-F Batch-2

Slide 1 — Title

Burmese Automatic Speech Recognition Using Kaldi

- Group members
- Course / Assignment
- Instructor
- Date

Subtitle:
From GMM-HMM to Convolutional-LSTM Acoustic Modeling

Slide 2 — Introduction

What is Automatic Speech Recognition?

- ASR converts **spoken language** → **text**
- Main components:
  - Acoustic Model
  - Pronunciation Lexicon
  - Language Model
  - Decoder
- Our project focuses on **Burmese ASR**
- We implemented both:
  - Traditional **GMM-HMM**
  - Neural **DNN-based** approaches

Slide 3 — Project Objectives

Our Objectives

1. Collect and prepare a Burmese speech dataset
2. Build a Kaldi-based ASR pipeline
3. Extract MFCC features
4. Create a Burmese pronunciation lexicon
5. Build Unigram, Bigram and Trigram LMs
6. Train GMM-HMM acoustic models
7. Test speaker adaptation with fMLLR

8. Experiment with **DNN-based acoustic models**
9. Compare all systems using WER and SER
10. Identify the best-performing model

⭐ Slide 4 — Data Collection

How We Collected the Data

Data Collection Process

Prepare Burmese Sentences

↓

Assign Sentences to Speakers

↓

Speakers Record Their Voices

↓

Collect Audio Recordings

↓

Match Audio with Transcriptions

↓

Organize Kaldi Data Files

Dataset Characteristics

- Burmese speech recordings
- Short command/number-oriented sentences
- Approximately **2 seconds per utterance**
- Recorded as **16 kHz, mono, 16-bit PCM**
- Each recording paired with its corresponding transcription
- Data divided into:
  - **Training set: 2,999 utterances**
  - **Testing set: 1,502 utterances**

Important point to explain

We collected speech recordings from multiple speakers so that the system could learn the relationship between Burmese speech sounds and their corresponding text.

Slide 5 — Data Preparation & Validation

Kaldi Data Files

```
data/train/
├── wav.scp
├── text
├── utt2spk
└── spk2utt

data/test/
├── wav.scp
├── text
├── utt2spk
└── spk2utt
```

Processing

- Checked audio files
- Removed missing recordings
- Matched audio with transcripts
- Created speaker mappings
- Validated Kaldi data directories

Final dataset:

2,999 training + 1,502 testing utterances

Slide 6 — Overall System Pipeline

Burmese Speech Data

↓

Data Collection

↓

Data Preparation

↓

MFCC Feature Extraction

↓

Pronunciation Lexicon

↓

Language Model

(Unigram / Bigram / Trigram)

↓

Acoustic Modeling

```
┌──────────────────┐
│ GMM-HMM     │
│ DNN       │
│ Conv + LSTM  │
└──────────────────┘
```

↓

HCLG / Decoding

↓

Speech Recognition

↓

WER / SER

↓

Model Comparison

This should be one of your **main visual slides**.

Slide 7 — MFCC Feature Extraction

Acoustic Features

- Sampling frequency: **16 kHz**
- 23 Mel filter banks
- 13 MFCC coefficients
- CMVN applied
- High-resolution features were also prepared for DNN experiments

Why MFCC?

MFCC features represent important characteristics of human speech and are widely used as acoustic features in speech recognition.

Slide 8 — Pronunciation Lexicon & Language Models

Pronunciation Lexicon

- Character-based Burmese phone representation
- **48 Burmese character units + SPN**
- Silence phone: `SIL`
- Vocabulary: **136 words**

Language Models

Unigram

↓

Bigram

↓

Trigram

The language model helps the decoder determine which word sequences are more likely.

Slide 9 — Traditional Acoustic Models

We gradually increased the complexity of the GMM-HMM system.

Tri1

Baseline GMM-HMM

↓

Tri2

LDA + MLLT

↓

Tri3

Speaker Adaptive Training (SAT)

Then we tested:

Tri3 + fMLLR decoding

This gives a clear experimental story:

> **Tri1 → Tri2 → Tri3 → fMLLR**

Slide 10 — DNN Acoustic Modeling

⭐ **Important new slide**

After the traditional GMM-HMM experiments, we experimented with **DNN-based acoustic modeling**.

> ### Neural Architecture
>
> Input MFCC Features
>
> ↓
>
> Convolutional Layers
>
> ↓
>
> LSTM Layers

↓

Dense / Output Layer

↓

Speech Recognition

Why DNN?

Traditional GMM-HMM models have limitations in learning complex speech patterns.

DNNs can learn more complex nonlinear relationships between:

Acoustic Features → Speech Units

Slide 11 — Experimental Strategy

We tested the systems progressively:

```
              Acoustic Model
                   |
        ┌──────────┴──────────┐
        ↓               ↓
     GMM-HMM              DNN
        |               |
   ┌────┼────┐                Conv + LSTM
   ↓    ↓    ↓               |
  Tri1  Tri2  Tri3               ↓
   |    |    |          WER ≈ 88%
   └────┴────┘
        |
     Bigram
        ↓
     Trigram
        ↓
```

fMLLR

Testing principle:

- Same test set
- Same reference transcripts
- Decode each system
- Calculate WER/SER
- Compare error types

Slide 12 — GMM-HMM Results

System	WER	SER
Tri1 + Bigram	124.46%	92.01%
Tri2 + Bigram	103.57%	95.14%
Tri3 + Bigram	**96.92%**	95.21%
Tri3 + Trigram	**96.06%**	**94.87%**
Tri3 + fMLLR + Trigram	104.39%	92.48%

Observation

Tri3 + Trigram was the best traditional GMM-HMM configuration.

Slide 13 — DNN Results

Neural Model Performance

Convolution + LSTM DNN

WER ≈ 88%

Compare:

Tri3 + Trigram

96.06%

↓

Conv + LSTM DNN

~88%

Improvement

The DNN system reduced WER by approximately **8 percentage points** compared with the best completed GMM-HMM system.

This is a very important result for your presentation.

Slide 14 — Overall Model Comparison

I recommend making this your **main results/competition slide**.

Model	WER	Key Observation
Tri1 + Bigram	124.46%	Baseline
Tri2 + Bigram	103.57%	LDA + MLLT improved performance
Tri3 + Bigram	96.92%	SAT further improved WER
Tri3 + Trigram	**96.06%**	Best GMM-HMM
Tri3 + fMLLR + Trigram	104.39%	Performance decreased
Conv + LSTM DNN	**~88%**	**Best overall** 🏆

Main conclusion

- **The Conv + LSTM DNN achieved the best performance with approximately 88% WER.**

Slide 15 — Analysis & Findings

Finding 1 — Acoustic Modeling

Tri1 ⟶ Tri2 ⟶ Tri3 consistently improved WER.

124.46% → 103.57% → 96.92%

Finding 2 — Language Modeling

Bigram provided a major improvement compared with the unigram baseline.

Trigram gave only a small additional improvement.

Finding 3 — fMLLR

fMLLR:

- Reduced deletion errors
- Increased substitution errors
- Increased insertion errors
- Overall WER became worse

Finding 4 — DNN

Conv + LSTM achieved ~88% WER, outperforming the traditional GMM-HMM systems.

This suggests that the neural model was better able to learn complex acoustic patterns in this dataset.

Slide 16 — Challenges, Future Work & Conclusion

You can combine these if you need to keep the presentation short.

Challenges

- Limited amount of Burmese speech data
- Short command-style recordings
- Limited vocabulary
- Burmese linguistic/Unicode complexity
- High recognition error rate

Future Work

- Collect more Burmese speech data
- Increase speaker and speaking-style diversity
- Improve pronunciation lexicon
- Improve language-model smoothing
- Experiment with stronger CNN/LSTM architectures
- Explore modern end-to-end models such as Transformer-based ASR

Conclusion

We successfully developed and evaluated a Burmese ASR system using Kaldi.
We compared traditional GMM-HMM models with neural acoustic modeling.
The best traditional system, **Tri3 + Trigram**, achieved **96.06% WER**, while the **Convolution + LSTM DNN achieved approximately 88% WER**, giving the best overall performance in our experiments.

🎯 Recommended Presentation Story

The strongest way to present your project is **not** to present the models as a random list. Tell it as an evolution:

1. We collected Burmese speech

↓

2. We prepared the dataset for Kaldi

↓

3. We built a traditional ASR baseline

↓

4. We improved the GMM-HMM

Tri1 → Tri2 → Tri3

↓

5. We experimented with language models

Bigram → Trigram

↓

6. We tested speaker adaptation

fMLLR

↓

7. We moved to neural acoustic modeling

Convolution + LSTM

↓

8. WER improved to ~88%

↓

9. Compare all approaches

↓

10. Discuss limitations & future work

⭐ The three numbers your audience should remember

- **124.46%** → GMM-HMM baseline
- **96.06%** → Best traditional GMM-HMM
- **~88% → Best Conv + LSTM DNN** 🏆

One important presentation tip: because the DNN result is approximately **88%**, use the exact WER from your DNN evaluation output in the final slides rather than writing more decimal places than your experiment actually produced.